

Musculoskeletal, Skin, and Connective Tissue

“Rigid, the skeleton of habit alone upholds the human frame.”
—Virginia Woolf

“Beauty may be skin deep, but ugly goes clear to the bone.”
—Redd Foxx

“The function of muscle is to pull and not to push, except in the case of the genitals and the tongue.”
—Leonardo da Vinci

“To thrive in life you need three bones. A wishbone. A backbone. And a funny bone.”
—Reba McEntire

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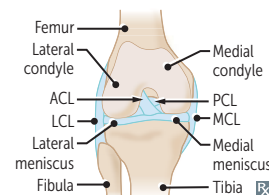
► MUSCULOSKELETAL, SKIN, AND CONNECTIVE TISSUE—ANATOMY AND PHYSIOLOGY

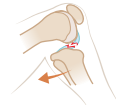
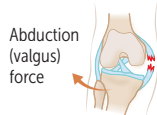
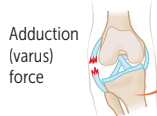
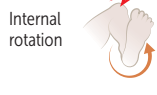
Knee exam

ACL: extends from lateral femoral condyle to anterior tibia.

PCL: extends from medial femoral condyle to posterior tibia.

Perform knee exam with patient supine.



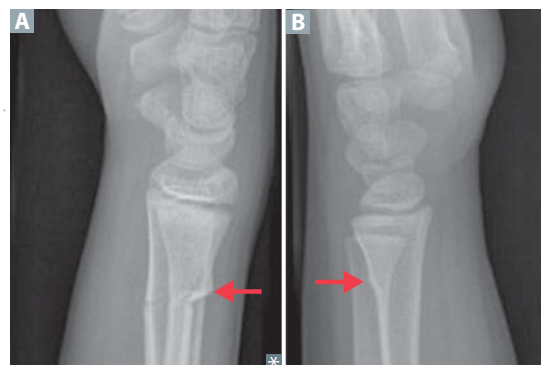
TEST	PROCEDURE	
Anterior drawer sign	Bending knee at 90° angle, ↑ anterior gliding of tibia due to ACL injury. Lachman test is similar, but at 30° angle.	 ACL tear Anterior drawer sign
Posterior drawer sign	Bending knee at 90° angle, ↑ posterior gliding of tibia due to PCL injury.	 PCL tear Posterior drawer sign
Abnormal passive abduction	Knee either extended or at ~ 30° angle, lateral (valgus) force → medial space widening of tibia → MCL injury.	 Abduction (valgus) force MCL tear
Abnormal passive adduction	Knee either extended or at ~ 30° angle, medial (varus) force → lateral space widening of tibia → LCL injury.	 Adduction (varus) force LCL tear
McMurray test	During flexion and extension of knee with rotation of tibia/foot: ▪ Pain, “popping” on external rotation → medial meniscal tear ▪ Pain, “popping” on internal rotation → lateral meniscal tear	 External rotation Medial tear  Internal rotation Lateral tear

Common pediatric fractures**Greenstick fracture**

Incomplete fracture extending partway through width of bone **A** following bending stress; bone is bent like a **green twig**.

Torus fracture

Axial force applied to immature bone → simple buckle fracture of cortex **B**. Can be very subtle.



Common knee conditions

“Unhappy triad”

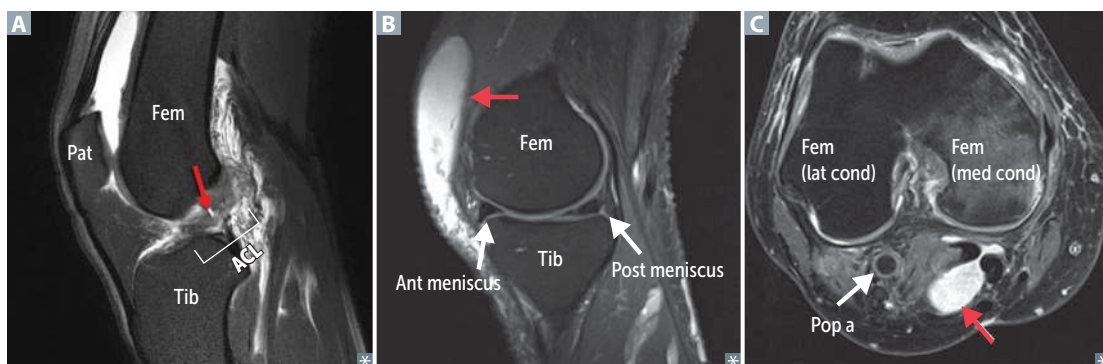
Common injury in contact sports due to lateral force applied to a planted leg. Classically, consists of damage to the ACL **A**, MCL, and medial meniscus (attached to MCL); however, lateral meniscus injury is more common. Presents with acute knee pain and signs of joint injury/instability.

Prepatellar bursitis

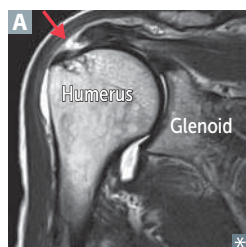
Inflammation of the prepatellar bursa in front of the kneecap **B**. Can be caused by repeated trauma or pressure from excessive kneeling (also called “housemaid’s knee”).

Baker cyst

Popliteal fluid collection (arrow in **C**) in gastrocnemius-semimembranosus bursa commonly communicating with synovial space and related to chronic joint disease.



Rotator cuff muscles

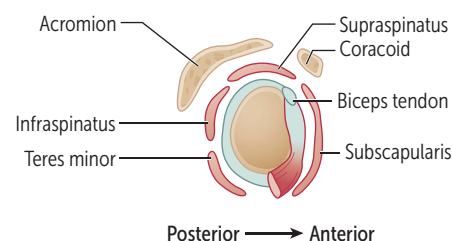


Shoulder muscles that form the rotator cuff:

- **S**upraspinatus (suprascapular nerve)—abducts arm initially (before the action of the deltoid); most common rotator cuff injury (trauma or degeneration and impingement → tendinopathy or tear [arrow in **A**]), assessed by “empty/full can” test.
- **I**nfraspinatus (suprascapular nerve)—laterally rotates arm; pitching injury.
- **t**eres minor (axillary nerve)—adducts and laterally rotates arm.
- **S**ubscapularis (upper and lower subscapular nerves)—medially rotates and adducts arm.

Innervated primarily by C5-C6.

SIItS (small t is for teres minor).



Arm abduction

DEGREE	MUSCLE	NERVE
0°–15°	Supraspinatus	Suprascapular
15°–100°	Deltoid	Axillary
> 90°	Trapezius	Accessory
> 100°	Serratus anterior	Long thoracic

Overuse injuries of the elbow

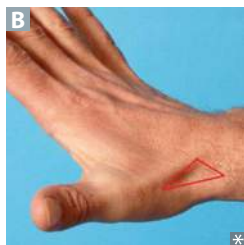
Medial epicondylitis (golfer's elbow)

Repetitive flexion (forehand shots) or idiopathic → pain near medial epicondyle.

Lateral epicondylitis (tennis elbow)

Repetitive extension (backhand shots) or idiopathic → pain near lateral epicondyle.

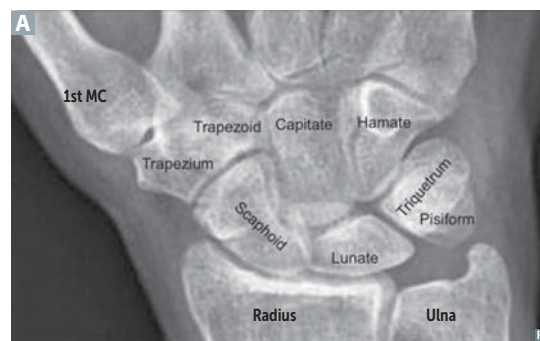
Wrist bones



Scaphoid, **L**unate, **T**riquetrum, **P**isiform, **H**amate, **C**apitate, **T**rapezoid, **T**rapezium **A**. (**S**o **L**ong **T**o **P**inky, **H**ere **C**omes **T**he **T**humb).

Scaphoid (palpable in anatomic snuff box **B**) is the most commonly fractured carpal bone, typically due to a fall on an outstretched hand. Complications of proximal scaphoid fractures include avascular necrosis and nonunion due to retrograde blood supply.

Dislocation of lunate may cause acute carpal tunnel syndrome.



Carpal tunnel syndrome

Entrapment of median nerve in carpal tunnel; nerve compression → paresthesia, pain, and numbness in distribution of median nerve (thenar eminence atrophies but sensation spared, because palmar cutaneous branch enters the hand external to carpal tunnel). Associated with pregnancy (due to edema), rheumatoid arthritis, hypothyroidism, diabetes, acromegaly, dialysis-related amyloidosis; may be associated with repetitive use. Suggested by ⊕ Tinel sign (percussion of wrist causes tingling) and Phalen maneuver (90° flexion of wrist causes tingling).

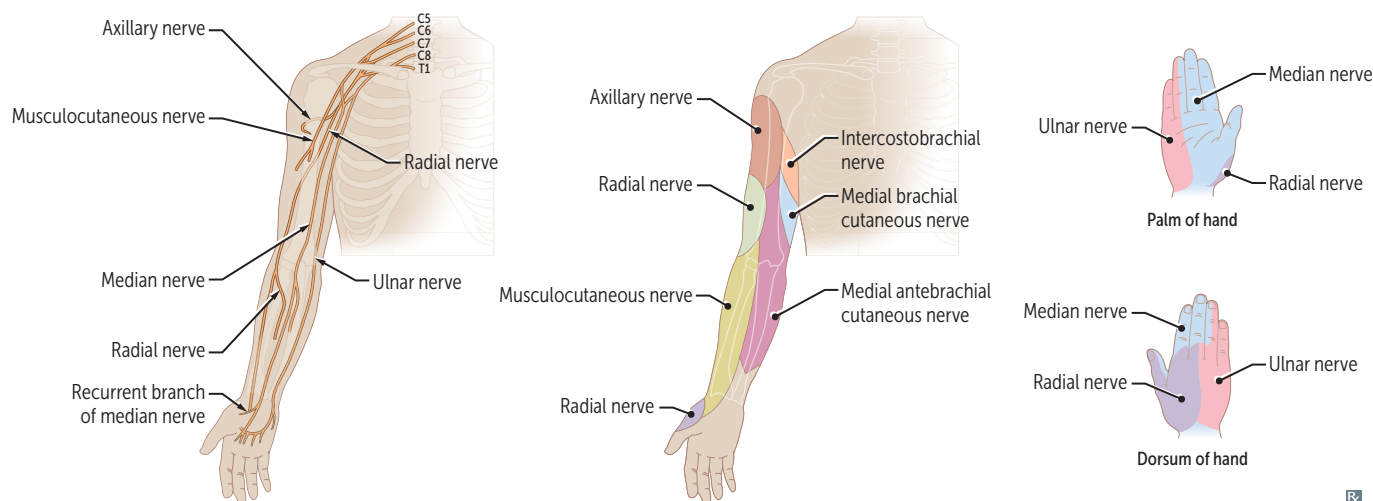
Guyon canal syndrome

Compression of ulnar nerve at wrist or hand. Classically seen in cyclists due to pressure from handlebars.

Upper extremity nerves

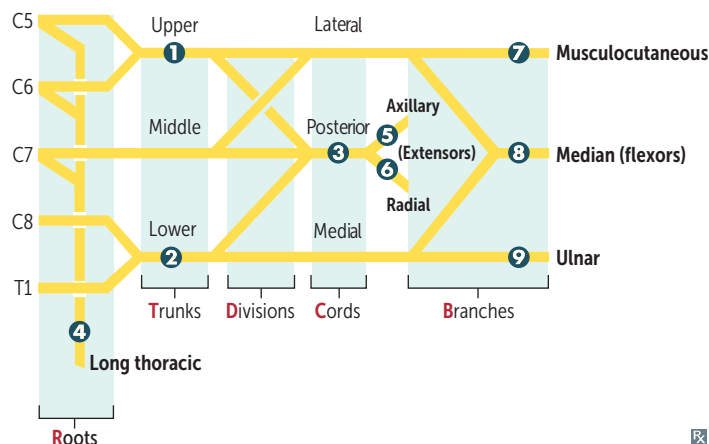
NERVE	CAUSES OF INJURY	PRESENTATION
Axillary (C5-C6)	Fractured surgical neck of humerus; anterior dislocation of humerus	Flattened deltoid Loss of arm abduction at shoulder ($> 15^\circ$) Loss of sensation over deltoid muscle and lateral arm
Musculocutaneous (C5-C7)	Upper trunk compression	Loss of forearm flexion and supination Loss of sensation over lateral forearm
Radial (C5-T1)	Midshaft fracture of humerus; compression of axilla, eg, due to crutches or sleeping with arm over chair ("Saturday night palsy")	Wrist drop: loss of elbow, wrist, and finger extension ↓ grip strength (wrist extension necessary for maximal action of flexors) Loss of sensation over posterior arm/forearm and dorsal hand
Median (C5-T1)	Supracondylar fracture of humerus (proximal lesion); carpal tunnel syndrome and wrist laceration (distal lesion)	"Ape hand" and "Pope's blessing" Loss of wrist flexion, flexion of lateral fingers, thumb opposition, lumbricals of 2nd and 3rd digits Loss of sensation over thenar eminence and dorsal and palmar aspects of lateral $3\frac{1}{2}$ fingers with proximal lesion
Ulnar (C8-T1)	Fracture of medial epicondyle of humerus "funny bone" (proximal lesion); fractured hook of hamate (distal lesion) from fall on outstretched hand	"Ulnar claw" on digit extension Radial deviation of wrist upon flexion (proximal lesion) Loss of wrist flexion, flexion of medial fingers, abduction and adduction of fingers (interossei), actions of medial 2 lumbrical muscles Loss of sensation over medial $1\frac{1}{2}$ fingers including hypothenar eminence
Recurrent branch of median nerve (C5-T1)	Superficial laceration of palm	"Ape hand" Loss of thenar muscle group: opposition, abduction, and flexion of thumb No loss of sensation

Humerus fractures, proximally to distally, follow the **ARM** (Axillary → Radial → Median)



Brachial plexus lesions

- 1 Erb palsy ("waiter's tip")
- 2 Klumpke palsy (claw hand)
- 3 Wrist drop
- 4 Winged scapula
- 5 Deltoid paralysis
- 6 "Saturday night palsy" (wrist drop)
- 7 Difficulty flexing elbow, variable sensory loss
- 8 Decreased thumb function, "Pope's blessing"
- 9 Intrinsic muscles of hand, claw hand



Randy
Travis
Drinks
Cold
Beer

CONDITION	INJURY	CAUSES	MUSCLE DEFICIT	FUNCTIONAL DEFICIT	PRESENTATION
Erb palsy ("waiter's tip")	Traction or tear of upper ("Erb-er") trunk: C5-C6 roots	Infants—lateral traction on neck during delivery Adults—trauma	Deltoid, supraspinatus Infraspinatus Biceps brachii	Abduction (arm hangs by side) Lateral rotation (arm medially rotated) Flexion, supination (arm extended and pronated)	
Klumpke palsy	Traction or tear of lower trunk: C8-T1 root	Infants—upward force on arm during delivery Adults—trauma (eg, grabbing a tree branch to break a fall)	Intrinsic hand muscles: lumbricals, interossei, thenar, hypothenar	Total claw hand: lumbricals normally flex MCP joints and extend DIP and PIP joints	
Thoracic outlet syndrome	Compression of lower trunk and subclavian vessels	Cervical rib (arrows in A), Pancoast tumor	Same as Klumpke palsy	Atrophy of intrinsic hand muscles; ischemia, pain, and edema due to vascular compression	
Winged scapula	Lesion of long thoracic nerve	Axillary node dissection after mastectomy, stab wounds	Serratus anterior	Inability to anchor scapula to thoracic cage → cannot abduct arm above horizontal position	

Distortions of the hand

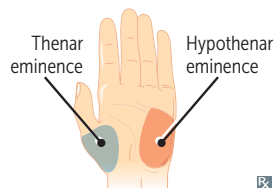
At rest, a balance exists between the extrinsic flexors and extensors of the hand, as well as the intrinsic muscles of the hand—particularly the lumbrical muscles (flexion of MCP, extension of DIP and PIP joints).

“Clawing”—seen best with **distal** lesions of median or ulnar nerves. Remaining extrinsic flexors of the digits exaggerate the loss of the lumbricals → fingers extend at MCP, flex at DIP and PIP joints.

Deficits less pronounced in **proximal** lesions; deficits present during voluntary flexion of the digits.

PRESENTATION				
CONTEXT	Extending fingers/at rest	Making a fist	Extending fingers/at rest	Making a fist
LOCATION OF LESION	Distal ulnar nerve	Proximal median nerve	Distal median nerve	Proximal ulnar nerve
SIGN	“Ulnar claw”	“Pope’s blessing”	“Median claw”	“OK gesture”

Note: Atrophy of the thenar eminence (unopposable thumb → “ape hand”) can be seen in median nerve lesions, while atrophy of the hypothenar eminence can be seen in ulnar nerve lesions.

Hand muscles

Thenar (median)—**O**pponens pollicis, **A**bductor pollicis brevis, **F**lexor pollicis brevis, superficial head (deep head by ulnar nerve).

Hypothenar (ulnar)—**O**pponens digiti minimi, **A**bductor digiti minimi, **F**lexor digiti minimi brevis.

Dorsal interossei—abduct the fingers.

Palmar interossei—adduct the fingers.

Lumbricals—flex at the MCP joint, extend PIP and DIP joints.

Both groups perform the same functions: **O**ppose, **A**bduct, and **F**lex (**OAF**).

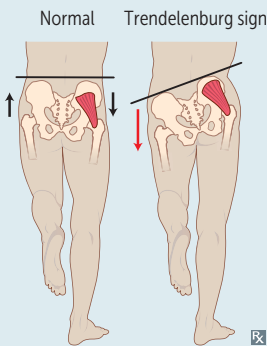
DAB = **D**orsals **AB**duct.

PAD = **P**almars **AD**duct.

Lower extremity nerves

NERVE	INNERVATION	CAUSE OF INJURY	PRESENTATION/COMMENTS
Iliohypogastric (T12-L1)	Sensory—suprapubic region Motor—transversus abdominis and internal oblique	Abdominal surgery	Burning or tingling pain in surgical incision site radiating to inguinal and suprapubic region
Genitofemoral nerve (L1-L2)	Sensory—scrotum/labia majora, medial thigh Motor—cremaster	Laparoscopic surgery	↓ anterior thigh sensation beneath inguinal ligament; absent cremasteric reflex
Lateral femoral cutaneous (L2-L3)	Sensory—anterior and lateral thigh	Tight clothing, obesity, pregnancy	↓ thigh sensation (anterior and lateral)
Obturator (L2-L4)	Sensory—medial thigh Motor—obturator externus, adductor longus, adductor brevis, gracilis, pectineus, adductor magnus	Pelvic surgery	↓ thigh sensation (medial) and adduction
Femoral (L2-L4)	Sensory—anterior thigh, medial leg Motor—quadriceps, iliopsoas, pectineus, sartorius	Pelvic fracture	↓ thigh flexion and leg extension
Sciatic (L4-S3)	Sensory—posterior thigh Motor—semitendinosus, semimembranosus, biceps femoris, adductor magnus	Herniated disc	Splits into common peroneal and tibial nerves
Common peroneal (L4-S2)	Sensory—dorsum of foot Motor—biceps femoris, tibialis anterior, extensor muscles of foot	Trauma or compression of lateral aspect of leg, fibular neck fracture	PED = P eroneal E verts and D orsiflexes; if injured, foot drop PED Loss of sensation on dorsum of foot Foot drop —inverted and plantarflexed at rest, loss of eversion and dorsiflexion; “steppage gait”
Tibial (L4-S3)	Sensory—sole of foot Motor—triceps surae, plantaris, popliteus, flexor muscles of foot	Knee trauma, Baker cyst (proximal lesion); tarsal tunnel syndrome (distal lesion)	TIP = T ibial I nverts and P lantarflexes; if injured, can't stand on TIP toes Inability to curl toes and loss of sensation on sole; in proximal lesions, foot everted at rest with loss of inversion and plantarflexion

Lower extremity nerves (continued)

NERVE	INNERVATION	CAUSE OF INJURY	PRESENTATION/COMMENTS
Superior gluteal (L4-S1) 	Motor—gluteus medius, gluteus minimus, tensor fascia latae	Iatrogenic injury during intramuscular injection to upper medial gluteal region	Trendelenburg sign/gait—pelvis tilts because weight-bearing leg cannot maintain alignment of pelvis through hip abduction Lesion is contralateral to the side of the hip that drops, ipsilateral to extremity on which the patient stands Choose superolateral quadrant (ideally the anterolateral region) as intramuscular injection site to avoid nerve injury
Inferior gluteal (L5-S2)	Motor—gluteus maximus	Posterior hip dislocation	Difficulty climbing stairs, rising from seated position; loss of hip extension
Pudendal (S2-S4)	Sensory—perineum Motor—external urethral and anal sphincters	Stretch injury during childbirth	↓ sensation in perineum and genital area; can cause fecal or urinary incontinence Can be blocked with local anesthetic during childbirth using ischial spine as a landmark for injection

Actions of hip muscles

ACTION	MUSCLES
Abductors	Gluteus medius, gluteus minimus
Adductors	Adductor magnus, adductor longus, adductor brevis
Extensors	Gluteus maximus, semitendinosus, semimembranosus
Flexors	Iliopsoas, rectus femoris, tensor fascia lata, pectineus
Internal rotation	Gluteus medius, gluteus minimus, tensor fascia latae
External rotation	Iliopsoas, gluteus maximus, piriformis, obturator

Signs of lumbosacral radiculopathy

Paresthesia and weakness related to specific lumbosacral spinal nerves. Usually, the intervertebral disc herniates into the central canal, affecting the inferior nerves (eg, herniation of L3/4 disc affects L4 spinal nerve, but not L3).

Intervertebral discs generally herniate posterolaterally, due to the thin posterior longitudinal ligament and thicker anterior longitudinal ligament along the midline of the vertebral bodies.

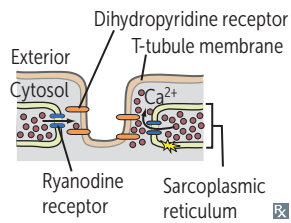
DISC LEVEL	FINDINGS
L3–L4	Weakness of knee extension, ↓ patellar reflex
L4–L5	Weakness of dorsiflexion, difficulty in heel-walking
L5–S1	Weakness of plantar flexion, difficulty in toe-walking, ↓ Achilles reflex

Neurovascular pairing

Nerves and arteries are frequently named together by the bones/regions with which they are associated. The following are exceptions to this naming convention.

LOCATION	NERVE	ARTERY
Axilla/lateral thorax	Long thoracic	Lateral thoracic
Surgical neck of humerus	Axillary	Posterior circumflex
Midshaft of humerus	Radial	Deep brachial
Distal humerus/ cubital fossa	Median	Brachial
Popliteal fossa	Tibial	Popliteal
Posterior to medial malleolus	Tibial	Posterior tibial

Muscle conduction to contraction

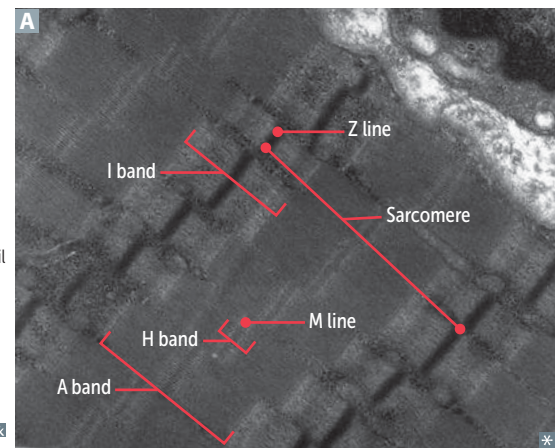
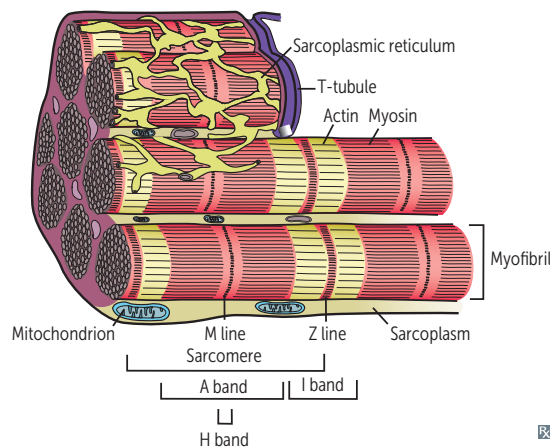


T-tubules are extensions of plasma membrane juxtaposed with terminal cisternae of the sarcoplasmic reticulum, allowing for coordinated contraction of muscles.

In skeletal muscle, 1 T-tubule + 2 terminal cisternae = triad.

In cardiac muscle, 1 T-tubule + 1 terminal cisterna = dyad.

1. Action potential depolarization opens presynaptic voltage-gated Ca^{2+} channels, inducing neurotransmitter release.
2. Postsynaptic ligand binding leads to muscle cell depolarization in the motor end plate.
3. Depolarization travels along muscle cell and down the T-tubule.
4. Depolarization of the voltage-sensitive dihydropyridine receptor, mechanically coupled to the ryanodine receptor on the sarcoplasmic reticulum, induces a conformational change in both receptors, causing Ca^{2+} release from sarcoplasmic reticulum.
5. Released Ca^{2+} binds to troponin C, causing a conformational change that moves tropomyosin out of the myosin-binding groove on actin filaments.
6. Myosin releases bound ADP and P_i → displacement of myosin on the actin filament (power stroke). Contraction results in shortening of **H** and **I** bands and between **Z** lines (**HIZ** shrinkage), but the **A** band remains the same length (**A** band is **A**lways the same length) **A**.
7. Binding of a new ATP molecule causes detachment of myosin head from actin filament. Hydrolysis of bound ATP → ADP, myosin head adopts high-energy position (“cocked”) for the next contraction cycle.



Types of muscle fibers

Type 1 muscle

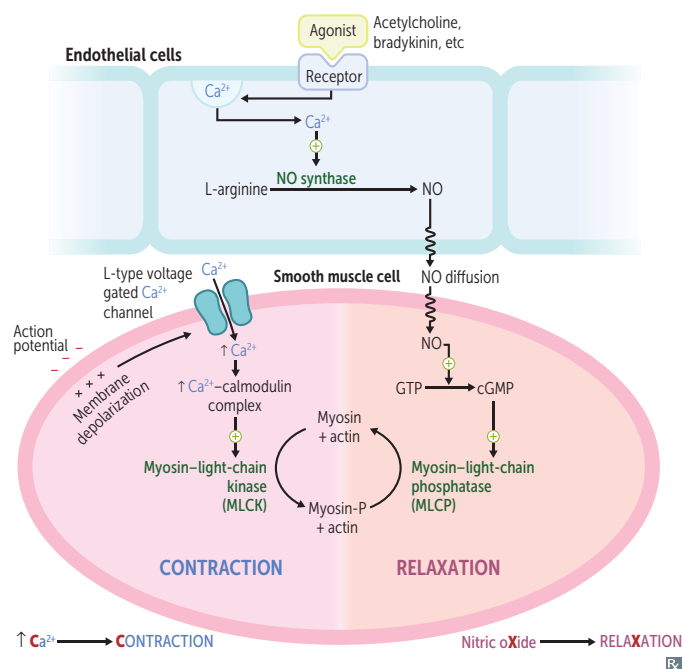
Slow twitch; **red** fibers resulting from ↑ mitochondria and myoglobin concentration (↑ **ox**idative phosphorylation) → sustained contraction. Proportion ↑ after endurance training.

Think “**1 slow red ox.**”

Type 2 muscle

Fast twitch; **white** fibers resulting from ↓ mitochondria and myoglobin concentration (↑ **anaerobic glycolysis**). Proportion ↑ after weight/resistance training, sprinting.

Smooth muscle contraction



Bone formation

Endochondral ossification

Bones of axial skeleton, appendicular skeleton, and base of skull. Cartilaginous model of bone is first made by chondrocytes. Osteoclasts and osteoblasts later replace with woven bone and then remodel to lamellar bone. In adults, woven bone occurs after fractures and in Paget disease. Defective in achondroplasia.

Membranous ossification

Bones of calvarium, facial bones, and clavicle. Woven bone formed directly without cartilage. Later remodeled to lamellar bone.

Cell biology of bone

Osteoblast

Builds **b**one by secreting collagen and catalyzing mineralization in alkaline environment via ALP. Differentiates from mesenchymal stem cells in periosteum. Osteoblastic activity measured by bone ALP, osteocalcin, propeptides of type I procollagen.

Osteoclast

Dissolves bone by secreting H^+ and collagenases. Differentiates from a fusion of monocyte/macrophage lineage precursors.

Parathyroid hormone

At low, intermittent levels, exerts anabolic effects (building bone) on osteoblasts and osteoclasts (indirect). Chronically ↑ PTH levels (1° hyperparathyroidism) cause catabolic effects (osteitis fibrosa cystica).

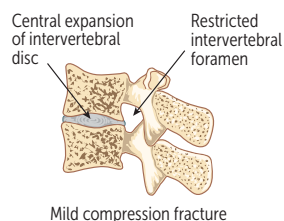
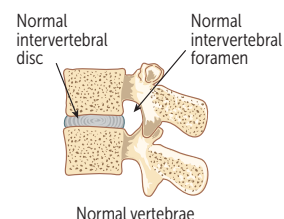
Estrogen

Inhibits apoptosis in bone-forming osteoblasts and induces apoptosis in bone-resorbing osteoclasts. Causes closure of epiphyseal plate during puberty. Estrogen deficiency (surgical or postmenopausal) → ↑ cycles of remodeling and bone resorption → ↑ risk of osteoporosis.

► MUSCULOSKELETAL, SKIN, AND CONNECTIVE TISSUE—PATHOLOGY

Achondroplasia

Failure of longitudinal bone growth (endochondral ossification) → short limbs. Membranous ossification is affected → large head relative to limbs. Constitutive activation of fibroblast growth factor receptor (FGFR3) actually inhibits chondrocyte proliferation. > 85% of mutations occur sporadically; autosomal dominant with full penetrance (homozygosity is lethal). Most common cause of dwarfism.

Osteoporosis

Trabecular (spongy) and cortical bone lose mass and interconnections despite normal bone mineralization and lab values (serum Ca^{2+} and PO_4^{3-}).

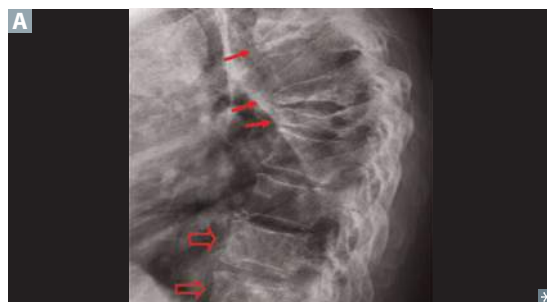
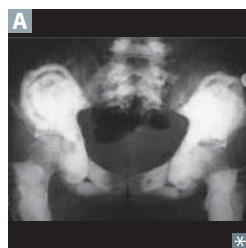
Most commonly due to ↑ bone resorption related to ↓ estrogen levels and old age. Can be secondary to drugs (eg, steroids, alcohol, anticonvulsants, anticoagulants, thyroid replacement therapy) or other medical conditions (eg, hyperparathyroidism, hyperthyroidism, multiple myeloma, malabsorption syndromes).

Diagnosed by a bone mineral density scan (dual-energy x-ray absorptiometry [DEXA]) with a T-score of ≤ -2.5 or by a fragility fracture of hip or vertebra. Screening recommended in women > 65 years old.

Prophylaxis: regular weight-bearing exercise and adequate Ca^{2+} and vitamin D intake throughout adulthood.

Treatment: bisphosphonates, teriparatide, SERMs, rarely calcitonin; denosumab (monoclonal antibody against RANKL).

Can lead to **vertebral compression fractures** (A), small arrows; large arrows show normal-for-age vertebral body height for comparison—acute back pain, loss of height, kyphosis. Also can present with fractures of femoral neck, distal radius (Colles fracture).

**Osteopetrosis**

Failure of normal bone resorption due to defective osteoclasts → thickened, dense bones that are prone to fracture. Defective osteoclasts cause overgrowth and sclerosis of cortical bone. Bone fills marrow space → pancytopenia, extramedullary hematopoiesis. Mutations (eg, carbonic anhydrase II) impair ability of osteoclast to generate acidic environment necessary for bone resorption. X-rays show diffuse symmetric skeletal sclerosis (bone-in-bone, “stone bone” A). Can result in cranial nerve impingement and palsies as a result of narrowed foramina. Bone marrow transplant is potentially curative as osteoclasts are derived from monocytes.

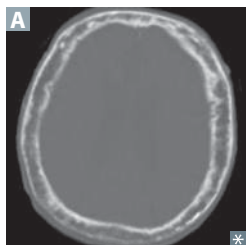
Osteomalacia/rickets

Defective mineralization of osteoid (osteomalacia) or cartilaginous growth plates (rickets, only in children). Most commonly due to vitamin D deficiency.

X-rays show osteopenia and “Looser zones” (pseudofractures) in osteomalacia, epiphyseal widening and metaphyseal cupping/fraying in rickets. Children with rickets have pathologic bow legs (genu varum **A**), bead-like costochondral junctions (rachitic rosary **B**), craniotabes (soft skull).

↓ vitamin D → ↓ serum Ca^{2+} → ↑ PTH secretion → ↓ serum PO_4^{3-} .

Hyperactivity of osteoblasts → ↑ ALP.

**Paget disease of bone (osteitis deformans)**

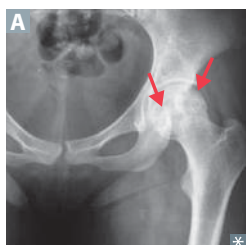
Common, localized disorder of bone remodeling caused by ↑ osteoclastic activity followed by ↑ osteoblastic activity that forms poor-quality bone. Serum Ca^{2+} , phosphorus, and PTH levels are normal. ↑ ALP. Mosaic pattern of woven and lamellar bone (osteocytes within lacunae in chaotic juxtapositions); long bone chalk-stick fractures. ↑ blood flow from ↑ arteriovenous shunts may cause high-output heart failure. ↑ risk of osteogenic sarcoma.

Hat size can be increased due to skull thickening **A**; hearing loss is common due to auditory foramen narrowing.

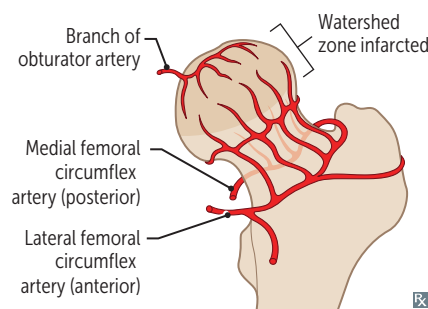
Stages of Paget disease:

- Lytic—osteoclasts
- Mixed—osteoclasts + osteoblasts
- Sclerotic—osteoblasts
- Quiescent—minimal osteoclast/osteoblast activity

Treatment: bisphosphonates.

Osteonecrosis (avascular necrosis)

Infarction of bone and marrow, usually very painful. Most common site is femoral head **A** (due to insufficiency of medial circumflex femoral artery). Causes include **C**orticosteroids, **A**lcoholism, **S**ickle cell disease, **T**rauma, “the **B**ends” (caisson/decompression disease), **L**Egg-Calvé-Perthes disease (idiopathic), **G**aucher disease, **S**lipped capital femoral epiphysis—**CAST** Bent **LEGS**.

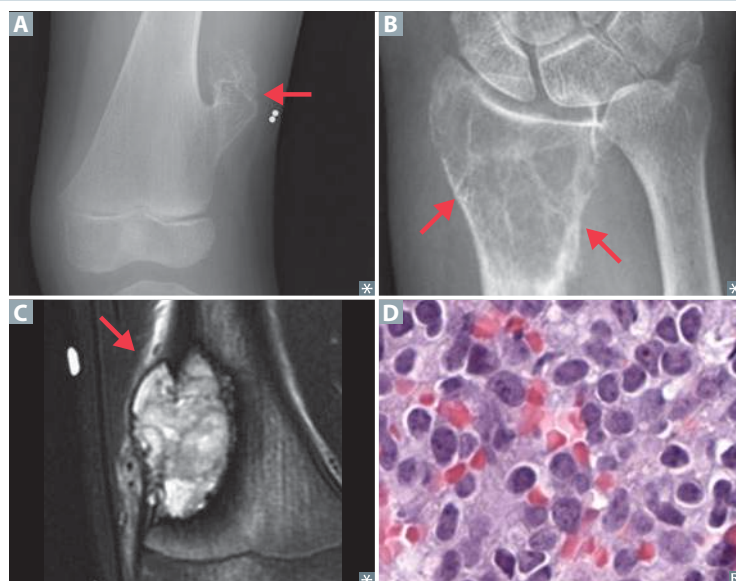
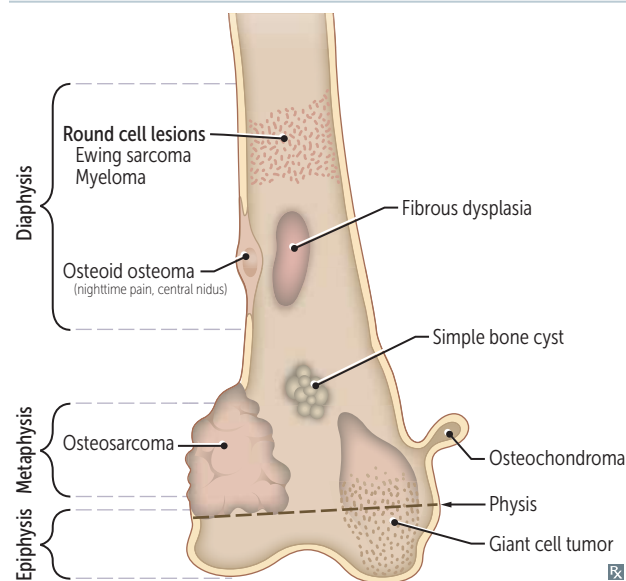


Lab values in bone disorders

DISORDER	SERUM Ca^{2+}	PO_4^{3-}	ALP	PTH	COMMENTS
Osteoporosis	—	—	—	—	↓ bone mass
Osteopetrosis	—/↓	—	—	—	Dense, brittle bones. Ca^{2+} ↓ in severe, malignant disease
Paget disease of bone	—	—	↑	—	Abnormal “mosaic” bone architecture
Osteitis fibrosa cystica					“Brown tumors” due to fibrous replacement of bone, subperiosteal thinning
Primary hyperparathyroidism	↑	↓	↑	↑	Idiopathic or parathyroid hyperplasia, adenoma, carcinoma
Secondary hyperparathyroidism	↓	↑	↑	↑	Often as compensation for CKD (↓ PO_4^{3-} excretion and production of activated vitamin D)
Osteomalacia/rickets	↓	↓	↑	↑	Soft bones; vitamin D deficiency also causes 2° hyperparathyroidism
Hypervitaminosis D	↑	↑	—	↓	Caused by oversupplementation or granulomatous disease (eg, sarcoidosis)

Primary bone tumors

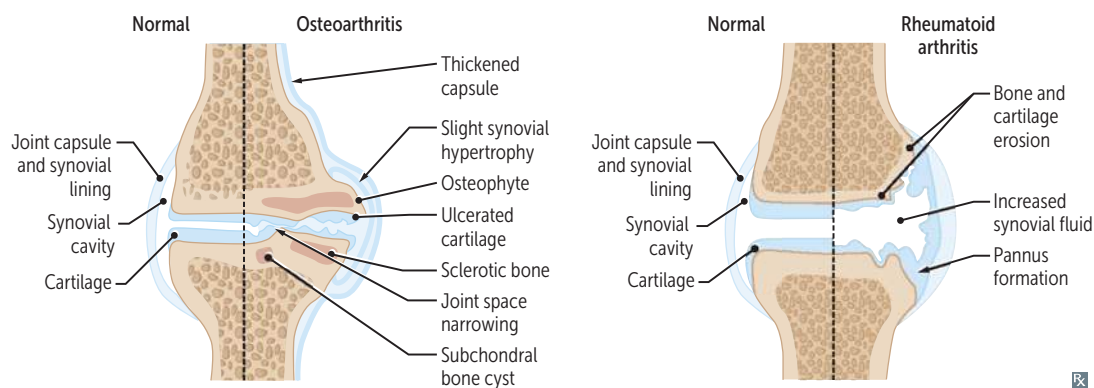
TUMOR TYPE	EPIDEMIOLOGY/LOCATION	CHARACTERISTICS
Benign tumors		
Osteochondroma	Most common benign bone tumor. Males < 25 years old.	Bony exostosis with cartilaginous (chondroid) cap A . Rarely transforms to chondrosarcoma.
Giant cell tumor	20–40 years old. Epiphysis of long bones (often in knee region). Arises most commonly at distal femur and proximal tibia. “Osteoclastoma.”	Locally aggressive benign tumor. “Soap bubble” appearance on x-ray B . Multinucleated giant cells that express RANKL.
Malignant tumors		
Osteosarcoma (osteogenic sarcoma)	One of the most common malignant bone tumors. Bimodal distribution: 10–20 years old (1°), > 65 (2°). Predisposing factors: Paget disease of bone, bone infarcts, radiation, familial retinoblastoma, Li-Fraumeni syndrome (germline <i>p53</i> mutation). Metaphysis of long bones, often around knee C .	Codman triangle (from elevation of periosteum) or sunburst pattern on x-ray. Aggressive. Treat with surgical en bloc resection (with limb salvage) and chemotherapy.
Ewing sarcoma	Boys < 15 years old. Commonly appears in diaphysis of long bones, pelvis, scapula, ribs.	Anaplastic small blue cell malignant tumor D . Extremely aggressive with early metastases, but responsive to chemotherapy. “Onion skin” periosteal reaction in bone. Associated with t(11;22) translocation causing fusion protein EWS-FLI 1. 11 + 22 = 33 (Patrick Ewing’s jersey number).



Osteoarthritis and rheumatoid arthritis

	Osteoarthritis	Rheumatoid arthritis
PATHOGENESIS	Mechanical—wear and tear destroys articular cartilage (degenerative joint disorder) → inflammation with inadequate repair. Chondrocytes mediate degradation and inadequate repair.	Autoimmune—inflammation induces formation of pannus (proliferative granulation tissue A), which erodes articular cartilage and bone.
PREDISPOSING FACTORS	Age, female, obesity, joint trauma.	Female, HLA-DR4, smoking, silica exposure. ⊕ rheumatoid factor (IgM antibody that targets IgG Fc region; in 80%), anti-cyclic citrullinated peptide antibody (more specific).
PRESENTATION	Pain in weight-bearing joints after use (eg, at the end of the day), improving with rest. Asymmetric joint involvement. Knee cartilage loss begins medially (“bowlegged”). No systemic symptoms.	Pain, swelling, and morning stiffness lasting > 1 hour, improving with use. Symmetric joint involvement. Systemic symptoms (fever, fatigue, weight loss). Extraarticular manifestations common.*
JOINT FINDINGS	Osteophytes (bone spurs), joint space narrowing, subchondral sclerosis and cysts. Synovial fluid non-inflammatory (WBC < 2000/mm ³). Involves DIP (Heberden nodes B) and PIP (Bouchard nodes C), and 1st CMC; not MCP.	Erosions, juxta-articular osteopenia, soft tissue swelling, subchondral cysts, joint space narrowing. Deformities: cervical subluxation, ulnar finger deviation, swan neck D , boutonniere E . Involves MCP, PIP, wrist; not DIP or 1st CMC. Synovial fluid inflammatory.
TREATMENT	Acetaminophen, NSAIDs, intra-articular glucocorticoids.	NSAIDs, glucocorticoids, disease-modifying agents (methotrexate, sulfasalazine, hydroxychloroquine, leflunomide), biologic agents (eg, TNF- α inhibitors).

*Extraarticular manifestations include rheumatoid nodules (fibrinoid necrosis with palisading histiocytes) in subcutaneous tissue and lung (+ pneumoconiosis → Caplan syndrome), interstitial lung disease, pleuritis, pericarditis, anemia of chronic disease, neutropenia + splenomegaly (Felty syndrome), AA amyloidosis, Sjögren syndrome, scleritis, carpal tunnel syndrome.



Gout

FINDINGS

Acute inflammatory monoarthritis caused by precipitation of monosodium urate crystals in joints **A**. More common in males. Associated with hyperuricemia, which can be caused by:

- Underexcretion of uric acid (90% of patients)—largely idiopathic, potentiated by renal failure; can be exacerbated by certain medications (eg, thiazide diuretics).
- Overproduction of uric acid (10% of patients)—Lesch-Nyhan syndrome, PRPP excess, ↑ cell turnover (eg, tumor lysis syndrome), von Gierke disease.

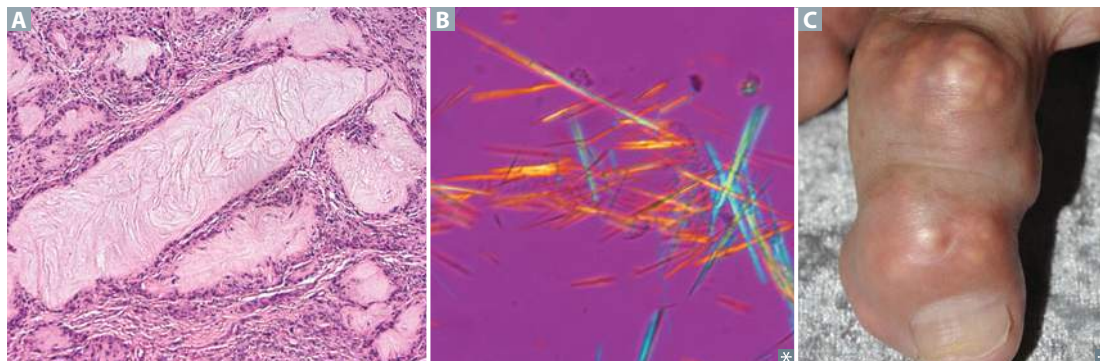
Crystals are needle shaped and $\ominus$ birefringent under polarized light (yellow under parallel light, blue under perpendicular light **B**).

SYMPTOMS

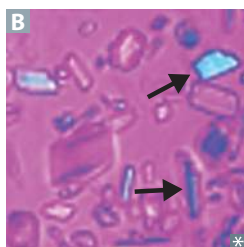
Asymmetric joint distribution. Joint is swollen, red, and painful. Classic manifestation is painful MTP joint of big toe (podagra). Tophus formation **C** (often on external ear, olecranon bursa, or Achilles tendon). Acute attack tends to occur after a large meal with foods rich in purines (eg, red meat, seafood), trauma, surgery, dehydration, diuresis, or alcohol consumption (alcohol metabolites compete for same excretion sites in kidney as uric acid → ↓ uric acid secretion and subsequent buildup in blood).

TREATMENT

Acute: NSAIDs (eg, indomethacin), glucocorticoids, colchicine.
Chronic (preventive): xanthine oxidase inhibitors (eg, allopurinol, febuxostat).



Calcium pyrophosphate deposition disease



Deposition of calcium pyrophosphate crystals within the joint space (arrows in **A**). Occurs in patients > 50 years old; both sexes affected equally. Usually idiopathic, sometimes associated with hemochromatosis, hyperparathyroidism, joint trauma.

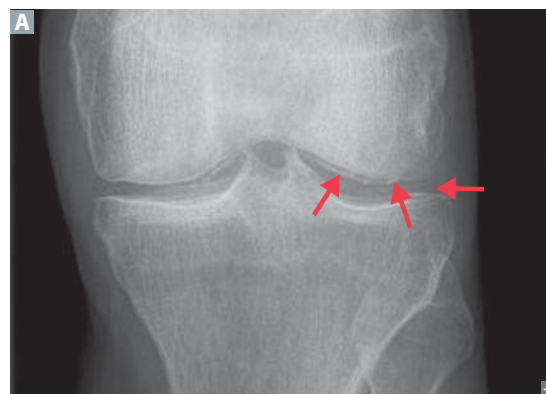
Pain and swelling with acute inflammation (pseudogout) and/or chronic degeneration (pseudo-osteoarthritis). Knee most commonly affected joint.

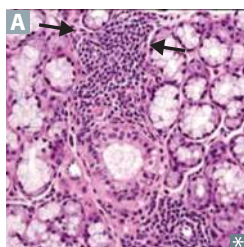
Chondrocalcinosis (cartilage calcification) on x-ray.

Crystals are rhomboid and weakly $\oplus$ birefringent under polarized light (blue when parallel to light) **B**.

Acute treatment: NSAIDs, colchicine, glucocorticoids.

Prophylaxis: colchicine.



Sjögren syndrome

Autoimmune disorder characterized by destruction of exocrine glands (especially lacrimal and salivary) by lymphocytic infiltrates **A**. Predominantly affects females 40–60 years old.

Findings:

- Inflammatory joint pain
- Keratoconjunctivitis sicca (↓ tear production and subsequent corneal damage)
- Xerostomia (↓ saliva production **B**)
- Presence of antinuclear antibodies, rheumatoid factor (can be in the absence of rheumatoid arthritis), antiribonucleoprotein antibodies: SS-A (anti-Ro) and/or SS-B (anti-La)
- Bilateral parotid enlargement

A common 1° disorder or a 2° syndrome associated with other autoimmune disorders (eg, rheumatoid arthritis, SLE, systemic sclerosis).

Complications: dental caries; mucosa-associated lymphoid tissue (MALT) lymphoma (may present as parotid enlargement).

Labial salivary gland biopsy can confirm diagnosis.

Septic arthritis

S aureus, *Streptococcus*, and *Neisseria gonorrhoeae* are common causes. Affected joint is swollen **A**, red, and painful. Synovial fluid purulent (WBC > 50,000/mm³).

Gonococcal arthritis—STI that presents as either purulent arthritis (eg, knee) or triad of polyarthralgia, tenosynovitis (eg, hand), dermatitis (eg, pustules).

**Seronegative
spondyloarthritis**

Arthritis without rheumatoid factor (no anti-IgG antibody). Strong association with HLA-B27 (MHC class I serotype). Subtypes (**PAIR**) share variable occurrence of inflammatory back pain (associated with morning stiffness, improves with exercise), peripheral arthritis, enthesitis (inflamed insertion sites of tendons, eg, Achilles), dactylitis (“sausage fingers”), uveitis.

Psoriatic arthritis

Associated with skin psoriasis and nail lesions. Asymmetric and patchy involvement **A**. Dactylitis and “pencil-in-cup” deformity of DIP on x-ray **B**.

Seen in fewer than 1/3 of patients with psoriasis.

**Ankylosing
spondylitis**

Symmetric involvement of spine and sacroiliac joints → ankylosis (joint fusion), uveitis, aortic regurgitation.

Bamboo spine (vertebral fusion) **C**. Can cause restrictive lung disease due to limited chest wall expansion (costovertebral and costosternal ankylosis).

More common in males.

**Inflammatory bowel
disease**

Crohn disease and ulcerative colitis are often associated with spondyloarthritis.

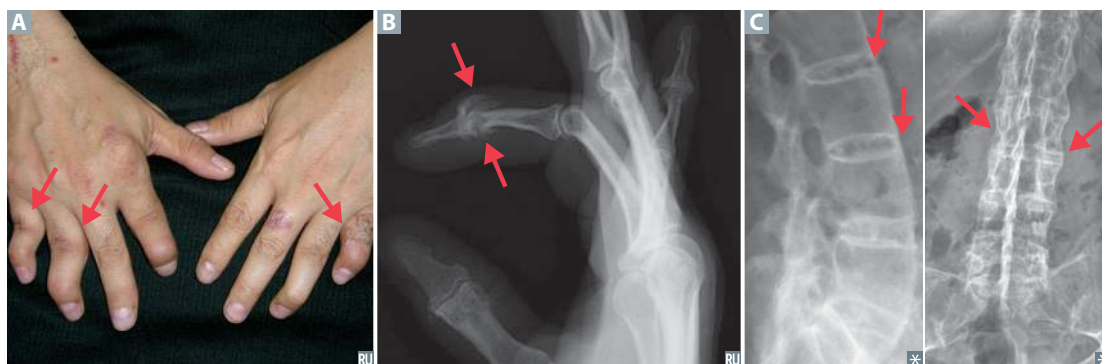
Reactive arthritis

Formerly known as Reiter syndrome. Classic triad:

- **Conjunctivitis**
- **Urethritis**
- **Arthritis**

“Can’t see, can’t pee, can’t bend my knee.”

Shigella, *Yersinia*, *Chlamydia*, *Campylobacter*, *Salmonella* (ShY ChiCS).



Systemic lupus erythematosus

SYMPTOMS



Classic presentation: rash, joint pain, and fever, most commonly in a female of reproductive age and African-American descent.

Libman-Sacks Endocarditis—nonbacterial, verrucous thrombi usually on mitral or aortic valve and can be present on either surface of the valve (but usually on undersurface) (**LSE** in **SLE**).

Lupus nephritis (glomerular deposition of anti-DNA immune complexes) can be nephritic or nephrotic (hematuria or proteinuria). Most common and severe type is diffuse proliferative.

Common causes of death in SLE:

- Cardiovascular disease
- Infections
- Renal disease

RASH OR PAIN:

Rash (malar **A** or discoid **B**)

Arthritis (nonerosive)

Serositis

Hematologic disorders (eg, cytopenias)

Oral/nasopharyngeal ulcers

Renal disease

Photosensitivity

Antinuclear antibodies

Immunologic disorder (anti-dsDNA, anti-Sm, antiphospholipid)

Neurologic disorders (eg, seizures, psychosis)

FINDINGS

Antinuclear antibodies (ANA)

Sensitive, not specific

Anti-dsDNA antibodies

Specific, poor prognosis (renal disease)

Anti-Smith antibodies

Specific, not prognostic (directed against snRNPs)

Antihistone antibodies

Sensitive for drug-induced lupus (eg, hydralazine, procainamide)

↓ C3, C4, and CH₅₀ due to immune complex formation.

TREATMENT

NSAIDs, steroids, immunosuppressants, hydroxychloroquine.

Antiphospholipid syndrome

1° or 2° autoimmune disorder (most commonly in SLE).

Diagnose based on clinical criteria including history of thrombosis (arterial or venous) or spontaneous abortion along with laboratory findings of lupus anticoagulant, anticardiolipin, anti-β₂ glycoprotein antibodies.

Treat with systemic anticoagulation.

Anticardiolipin antibodies can cause false-positive VDRL/RPR, and lupus anticoagulant can cause prolonged PTT, which is not corrected by the addition of normal platelet-free plasma.

Mixed connective tissue disease

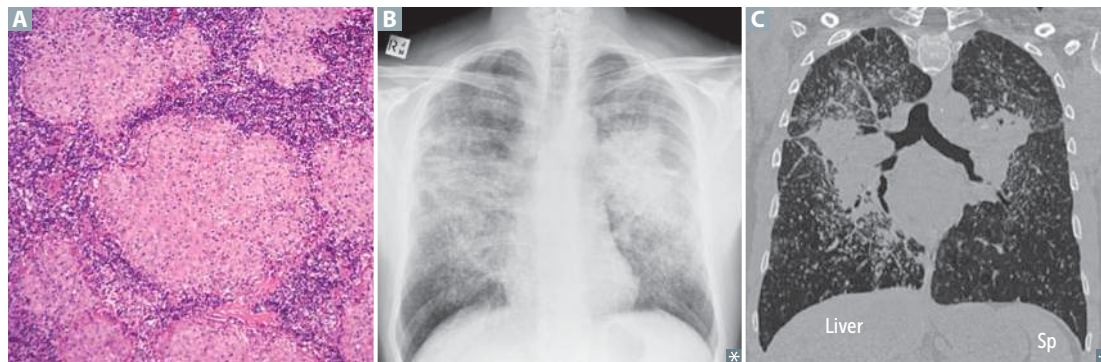
Features of SLE, systemic sclerosis, and/or polymyositis. Associated with anti-U1 RNP antibodies (speckled ANA).

Sarcoidosis

Characterized by immune-mediated, widespread noncaseating granulomas **A**, elevated serum ACE levels, and elevated CD4+/CD8+ ratio in bronchoalveolar lavage fluid. More common in African-American females. Often asymptomatic except for enlarged lymph nodes. Findings on CXR of bilateral adenopathy and coarse reticular opacities **B**; CT of the chest better demonstrates the extensive hilar and mediastinal adenopathy **C**.

Associated with restrictive lung disease (interstitial fibrosis), erythema nodosum, lupus pernio (skin lesions on face resembling lupus), Bell palsy, epithelioid granulomas containing microscopic Schaumann and asteroid bodies, uveitis, hypercalcemia (due to ↑ 1 α -hydroxylase-mediated vitamin D activation in macrophages).

Treatment: steroids (if symptomatic).



Polymyalgia rheumatica

SYMPTOMS	Pain and stiffness in shoulders and hips, often with fever, malaise, weight loss. Does not cause muscular weakness. More common in women > 50 years old; associated with giant cell (temporal) arteritis.
FINDINGS	↑ ESR, ↑ CRP, normal CK.
TREATMENT	Rapid response to low-dose corticosteroids.

Fibromyalgia

Most commonly seen in women 20–50 years old. Chronic, widespread musculoskeletal pain associated with “tender points,” stiffness, paresthesias, poor sleep, fatigue, cognitive disturbance (“fibro fog”). Treatment: regular exercise, antidepressants (TCAs, SNRIs), anticonvulsants.

**Polymyositis/
dermatomyositis**

↑ CK, ⊕ ANA, ⊕ anti-Jo-1, ⊕ anti-SRP, ⊕ anti-Mi-2 antibodies. Treatment: steroids followed by long-term immunosuppressant therapy (eg, methotrexate).

Polymyositis

Progressive symmetric proximal muscle weakness, characterized by endomysial inflammation with CD8+ T cells. Most often involves shoulders.

Dermatomyositis

Similar to polymyositis, but also involves malar rash (similar to SLE), Gottron papules **A**, heliotrope (erythematous periorbital) rash **B**, “shawl and face” rash **C**, “mechanic’s hands.” ↑ risk of occult malignancy. Perimysial inflammation and atrophy with CD4+ T cells.

**Neuromuscular junction diseases**

	Myasthenia gravis	Lambert-Eaton myasthenic syndrome
FREQUENCY	Most common NMJ disorder	Uncommon
PATHOPHYSIOLOGY	Autoantibodies to postsynaptic ACh receptor	Autoantibodies to presynaptic Ca ²⁺ channel → ↓ ACh release
CLINICAL	Ptosis, diplopia, weakness Worsens with muscle use Improvement after edrophonium (tensilon) test	Proximal muscle weakness, autonomic symptoms (dry mouth, impotence) Improves with muscle use
ASSOCIATED WITH	Thymoma, thymic hyperplasia	Small cell lung cancer
ACH E INHIBITOR ADMINISTRATION	Reverses symptoms (edrophonium to diagnose, pyridostigmine to treat)	Minimal effect

Raynaud phenomenon

↓ blood flow to skin due to arteriolar (small vessel) vasospasm in response to cold or stress: color change from white (ischemia) to blue (hypoxia) to red (reperfusion). Most often in the fingers **A** and toes. Called **Raynaud disease** when 1° (idiopathic), **Raynaud syndrome** when 2° to a disease process such as mixed connective tissue disease, SLE, or CREST syndrome (limited form of systemic sclerosis). Digital ulceration (critical ischemia) seen in 2° Raynaud syndrome. Treat with Ca²⁺ channel blockers.



Scleroderma (systemic sclerosis)

Triad of autoimmunity, noninflammatory vasculopathy, and collagen deposition with fibrosis. Commonly sclerosis of skin, manifesting as puffy, taut skin **A** without wrinkles, fingertip pitting **B**. Also sclerosis of renal, pulmonary (most common cause of death), cardiovascular, GI systems. 75% female. 2 major types:

- **Diffuse scleroderma**—widespread skin involvement, rapid progression, early visceral involvement. Associated with anti-Scl-70 antibody (anti-DNA topoisomerase I antibody).
- **Limited scleroderma**—limited skin involvement confined to fingers and face. Also with **CREST** syndrome: **C**alcinosis **C**, anti-**C**entromere antibody, **R**aynaud phenomenon, **E**sophageal dysmotility, **S**clerodactyly, and **T**elangiectasia. More benign clinical course.



► MUSCULOSKELETAL, SKIN, AND CONNECTIVE TISSUE—DERMATOLOGY

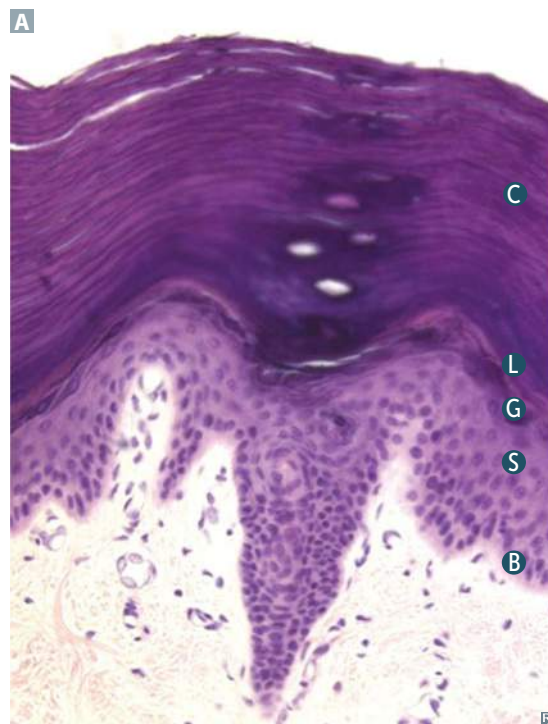
Skin layers

Skin has 3 layers: epidermis, dermis, subcutaneous fat (hypodermis, subcutis).

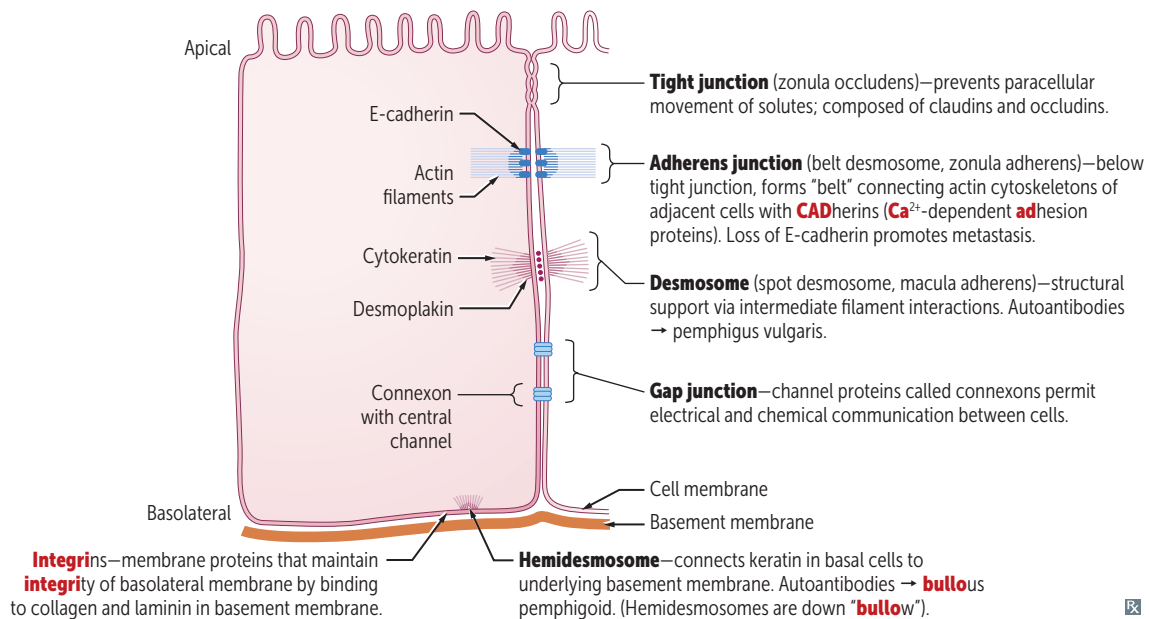
Epidermis layers from surface to base **A**:

- Stratum **C**orneum (keratin)
- Stratum **L**ucidum
- Stratum **G**ranulosum
- Stratum **S**pinosum (desmosomes)
- Stratum **B**asale (stem cell site)

Californians **L**ike **G**irls in **S**tring **B**ikinis.



Epithelial cell junctions



R

Dermatologic macroscopic terms (morphology)

LESION	CHARACTERISTICS	EXAMPLES
Macule	Flat lesion with well-circumscribed change in skin color < 1 cm	Freckle, labial macule A
Patch	Macule > 1 cm	Large birthmark (congenital nevus) B
Papule	Elevated solid skin lesion < 1 cm	Mole (nevus) C , acne
Plaque	Papule > 1 cm	Psoriasis D
Vesicle	Small fluid-containing blister < 1 cm	Chickenpox (varicella), shingles (zoster) E
Bulla	Large fluid-containing blister > 1 cm	Bullous pemphigoid F
Pustule	Vesicle containing pus	Pustular psoriasis G
Wheal	Transient smooth papule or plaque	Hives (urticaria) H
Scale	Flaking off of stratum corneum	Eczema, psoriasis, SCC I
Crust	Dry exudate	Impetigo J



Dermatologic microscopic terms

LESION	CHARACTERISTICS	EXAMPLES
Hyperkeratosis	↑ thickness of stratum corneum	Psoriasis, calluses
Parakeratosis	Hyperkeratosis with retention of nuclei in stratum corneum	Psoriasis
Hypergranulosis	↑ thickness of stratum granulosum	Lichen planus
Spongiosis	Epidermal accumulation of edematous fluid in intercellular spaces	Eczematous dermatitis
Acantholysis	Separation of epidermal cells	Pemphigus vulgaris
Acanthosis	Epidermal hyperplasia (↑ spinosum)	Acanthosis nigricans

Pigmented skin disorders

Albinism	Normal melanocyte number with ↓ melanin production A due to ↓ tyrosinase activity or defective tyrosine transport. ↑ risk of skin cancer.
Melasma (chloasma)	Hyperpigmentation associated with pregnancy (“mask of pregnancy” B) or OCP use.
Vitiligo	Irregular areas of complete depigmentation C . Caused by autoimmune destruction of melanocytes.



Common skin disorders

Acne	Multifactorial etiology—↑ sebum/androgen production, abnormal keratinocyte desquamation, <i>Propionibacterium acnes</i> colonization of the pilosebaceous unit (comedones), and inflammation (papules/pustules A , nodules, cysts). Treatment includes retinoids, benzoyl peroxide, and antibiotics.
Atopic dermatitis (eczema)	Pruritic eruption, commonly on skin flexures. Often associated with other atopic diseases (asthma, allergic rhinitis, food allergies); ↑ serum IgE. Usually appears on face in infancy B and then in antecubital fossa C in children and adults.
Allergic contact dermatitis	Type IV hypersensitivity reaction that follows exposure to allergen. Lesions occur at site of contact (eg, nickel D , poison ivy, neomycin E).
Melanocytic nevus	Common mole. Benign, but melanoma can arise in congenital or atypical moles. Intradermal nevi are papular F . Junctional nevi are flat macules G .
Pseudofolliculitis barbae	Foreign body inflammatory facial skin disorder characterized by firm, hyperpigmented papules and pustules that are painful and pruritic. Located on cheeks, jawline, and neck. Commonly occurs as a result of shaving (“razor bumps”), primarily affects African-American males.
Psoriasis	Papules and plaques with silvery scaling H , especially on knees and elbows. Acanthosis with parakeratotic scaling (nuclei still in stratum corneum), Munro microabscesses. ↑ stratum spinosum, ↓ stratum granulosum. Auspitz sign (arrow in I)—pinpoint bleeding spots from exposure of dermal papillae when scales are scraped off. Can be associated with nail pitting and psoriatic arthritis.
Rosacea	Inflammatory facial skin disorder characterized by erythematous papules and pustules J , but no comedones. May be associated with facial flushing in response to external stimuli (eg, alcohol, heat). Phymatous rosacea can cause rhinophyma (bulbous deformation of nose).
Seborrheic keratosis	Flat, greasy, pigmented squamous epithelial proliferation with keratin-filled cysts (horn cysts) K . Looks “stuck on.” Lesions occur on head, trunk, and extremities. Common benign neoplasm of older persons. Leser-Trélat sign L —sudden appearance of multiple seborrheic keratoses, indicating an underlying malignancy (eg, GI, lymphoid).
Verrucae	Warts; caused by HPV. Soft, tan-colored, cauliflower-like papules M . Epidermal hyperplasia, hyperkeratosis, koilocytosis. Condyloma acuminatum on genitals N .
Urticaria	Hives. Pruritic wheals that form after mast cell degranulation O . Characterized by superficial dermal edema and lymphatic channel dilation.



Vascular tumors of skin

Angiosarcoma	Rare blood vessel malignancy typically occurring in the head, neck, and breast areas. Usually in elderly, on sun-exposed areas. Associated with radiation therapy and chronic postmastectomy lymphedema. Hepatic angiosarcoma associated with vinyl chloride and arsenic exposures. Very aggressive and difficult to resect due to delay in diagnosis.
Bacillary angiomatosis	Benign capillary skin papules A found in AIDS patients. Caused by <i>Bartonella henselae</i> infections. Frequently mistaken for Kaposi sarcoma, but has neutrophilic infiltrate.
Cherry hemangioma	Benign capillary hemangioma of the elderly B . Does not regress. Frequency ↑ with age.
Cystic hygroma	Cavernous lymphangioma of the neck C . Associated with Turner syndrome.
Glomus tumor	Benign, painful, red-blue tumor, commonly under fingernails D . Arises from modified smooth muscle cells of the thermoregulatory glomus body.
Kaposi sarcoma	Endothelial malignancy most commonly of the skin, but also mouth, GI tract, and respiratory tract. Associated with HHV-8 and HIV. Rarely mistaken for bacillary angiomatosis, but has lymphocytic infiltrate.
Pyogenic granuloma	Polypoid lobulated capillary hemangioma E that can ulcerate and bleed. Associated with trauma and pregnancy.
Strawberry hemangioma	Benign capillary hemangioma of infancy F . Appears in first few weeks of life (1/200 births); grows rapidly and regresses spontaneously by 5–8 years old.



Skin infections**Bacterial infections****Impetigo**

Very superficial skin infection. Usually from *S aureus* or *S pyogenes*. Highly contagious. Honey-colored crusting **A**.

Bullous impetigo **B** has bullae and is usually caused by *S aureus*.

Erysipelas

Infection involving upper dermis and superficial lymphatics, usually from *S pyogenes*. Presents with well-defined demarcation between infected and normal skin **C**.

Cellulitis

Acute, painful, spreading infection of deeper dermis and subcutaneous tissues. Usually from *S pyogenes* or *S aureus*. Often starts with a break in skin from trauma or another infection **D**.

Abscess

Collection of pus from a walled-off infection within deeper layers of skin **E**. Offending organism is almost always *S aureus*.

Necrotizing fasciitis

Deeper tissue injury, usually from anaerobic bacteria or *S pyogenes*. Pain may be out of proportion to exam findings. Results in crepitus from methane and CO₂ production. “Flesh-eating bacteria.” Causes bullae and a purple color to the skin **F**.

Staphylococcal scalded skin syndrome

Exotoxin destroys keratinocyte attachments in stratum granulosum only (vs toxic epidermal necrolysis, which destroys epidermal-dermal junction). Characterized by fever and generalized erythematous rash with sloughing of the upper layers of the epidermis **G** that heals completely. ⊕ Nikolsky sign. Seen in newborns and children, adults with renal insufficiency.

Viral infections**Herpes**

Herpes virus infections (HSV1 and HSV2) of skin can occur anywhere from mucosal surfaces to normal skin. These include herpes labialis, herpes genitalis, herpetic whitlow **H** (finger).

Molluscum contagiosum

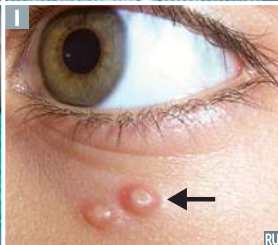
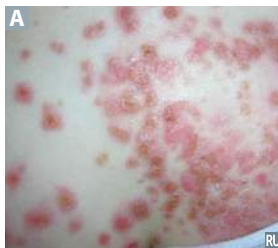
Umbilicated papules **I** caused by a poxvirus. While frequently seen in children, it may be sexually transmitted in adults.

Varicella zoster virus

Causes varicella (chickenpox) and zoster (shingles). Varicella presents with multiple crops of lesions in various stages from vesicles to crusts. Zoster is a reactivation of the virus in dermatomal distribution (unless it is disseminated).

Hairy leukoplakia

Irregular, white, painless plaques on lateral tongue that cannot be scraped off **J**. EBV mediated. Occurs in HIV-positive patients, organ transplant recipients. Contrast with thrush (scrapable) and leukoplakia (precancerous).



Blistering skin disorders**Pemphigus vulgaris**

Potentially fatal autoimmune skin disorder with IgG antibody against desmoglein (component of desmosomes, which connect keratinocytes in the stratum spinosum). Flaccid intraepidermal bullae **A** caused by acantholysis (separation of keratinocytes, resembling a “row of tombstones”); oral mucosa is also involved. Type II hypersensitivity reaction. Immunofluorescence reveals antibodies around epidermal cells in a reticular (net-like) pattern **B**. Nikolsky sign $\oplus$ (separation of epidermis upon manual stroking of skin).

Bullous pemphigoid

Less severe than pemphigus vulgaris. Involves IgG antibody against hemidesmosomes (epidermal basement membrane; antibodies are “**bullo**” the epidermis). Tense blisters **C** containing eosinophils affect skin but spare oral mucosa. Immunofluorescence reveals linear pattern at epidermal-dermal junction **D**. Nikolsky sign $\ominus$.

Dermatitis herpetiformis

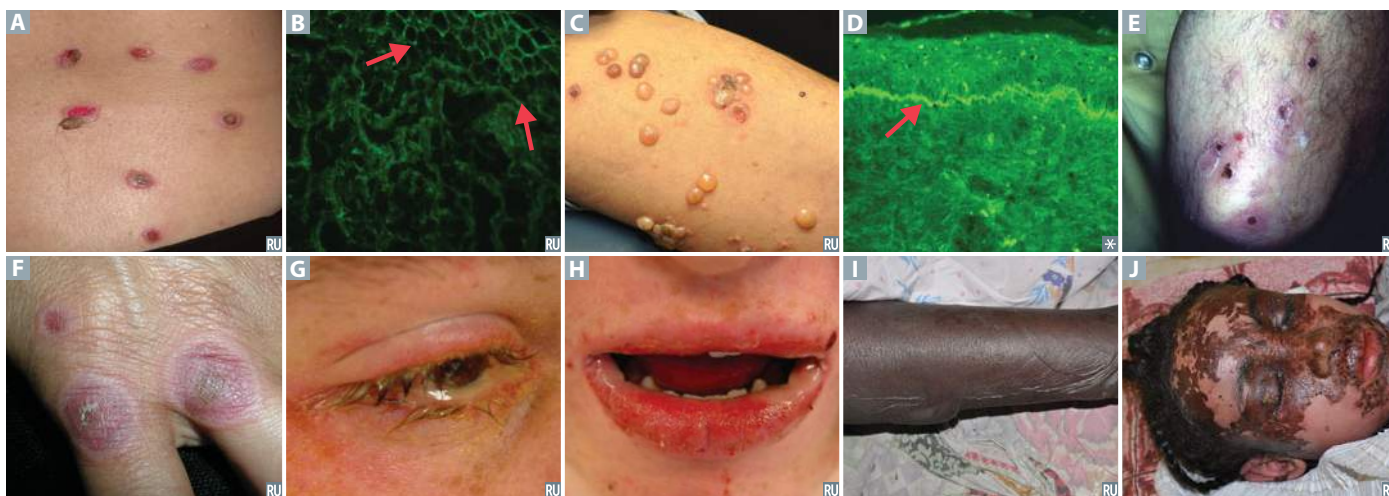
Pruritic papules, vesicles, and bullae (often found on elbows) **E**. Deposits of IgA at tips of dermal papillae. Associated with celiac disease. Treatment: dapsone, gluten-free diet.

Erythema multiforme

Associated with infections (eg, *Mycoplasma pneumoniae*, HSV), drugs (eg, sulfa drugs, β -lactams, phenytoin), cancers, autoimmune disease. Presents with multiple types of lesions—macules, papules, vesicles, target lesions (look like targets with multiple rings and dusky center showing epithelial disruption) **F**.

Stevens-Johnson syndrome

Characterized by fever, bullae formation and necrosis, sloughing of skin at dermal-epidermal junction, high mortality rate. Typically 2 mucous membranes are involved **G H**, and targetoid skin lesions may appear, as seen in erythema multiforme. Usually associated with adverse drug reaction. A more severe form of Stevens-Johnson syndrome (SJS) with $> 30\%$ of the body surface area involved is **toxic epidermal necrolysis I J (TEN)**. 10–30% involvement denotes SJS-TEN.



Miscellaneous skin disorders

Acanthosis nigricans	Epidermal hyperplasia causing symmetric, hyperpigmented thickening of skin, especially in axilla or on neck A B . Associated with insulin resistance (eg, diabetes, obesity, Cushing syndrome), visceral malignancy (eg, gastric adenocarcinoma).
Actinic keratosis	Premalignant lesions caused by sun exposure. Small, rough, erythematous or brownish papules or plaques C D . Risk of squamous cell carcinoma is proportional to degree of epithelial dysplasia.
Erythema nodosum	Painful, raised inflammatory lesions of subcutaneous fat (panniculitis), usually on anterior shins. Often idiopathic, but can be associated with sarcoidosis, coccidioidomycosis, histoplasmosis, TB, streptococcal infections E , leprosy F , inflammatory bowel disease.
Lichen Planus	P ruritic, P urple, P olygonal P lanar P apules and P laques are the 6 P 's of lichen P lanus G H . Mucosal involvement manifests as Wickham striae (reticular white lines) and hypergranulosis. Sawtooth infiltrate of lymphocytes at dermal-epidermal junction. Associated with hepatitis C.
Pityriasis rosea	"Herald patch" I followed days later by other scaly erythematous plaques, often in a "Christmas tree" distribution on trunk J . Multiple plaques with collarette scale. Self-resolving in 6–8 weeks.
Sunburn	Acute cutaneous inflammatory reaction due to excessive UV irradiation. Causes DNA mutations, inducing apoptosis of keratinocytes. UVB is dominant in sun B urn, UVA in t A nning and photo A ging. Exposure to UVA and UVB ↑ risk of skin cancer (basal cell carcinoma, squamous cell carcinoma, melanoma). Can also lead to impetigo.



Skin cancer

Basal cell carcinoma

Most common skin cancer. Found in sun-exposed areas of body (eg, face). Locally invasive, but rarely metastasizes. Waxy, pink, pearly nodules, commonly with telangiectasias, rolled borders, central crusting or ulceration **A**. BCCs also appear as nonhealing ulcers with infiltrating growth **B** or as a scaling plaque (superficial BCC) **C**. Basal cell tumors have “palisading” nuclei **D**.



Squamous cell carcinoma

Second most common skin cancer. Associated with excessive exposure to sunlight, immunosuppression, chronically draining sinuses, and occasionally arsenic exposure. Commonly appears on face **E**, lower lip **F**, ears, hands. Locally invasive, may spread to lymph nodes, and will rarely metastasize. Ulcerative red lesions with frequent scale. Histopathology: keratin “pearls” **G**.

Actinic keratosis, a scaly plaque, is a precursor to squamous cell carcinoma.

Keratoacanthoma is a variant that grows rapidly (4–6 weeks) and may regress spontaneously over months **H**.



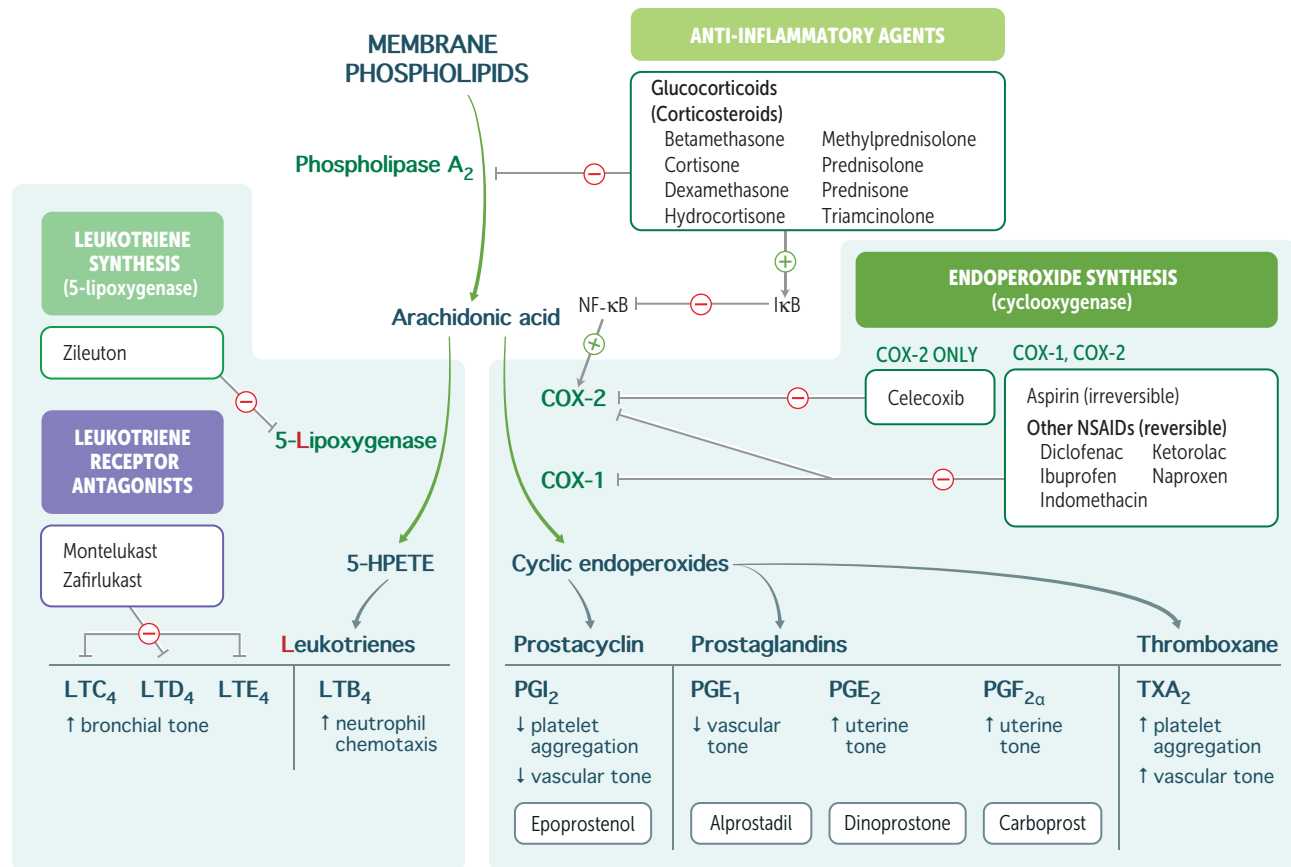
Melanoma

Common tumor with significant risk of metastasis. S-100 tumor marker. Associated with sunlight exposure and dysplastic nevi; fair-skinned persons are at ↑ risk. Depth of tumor (Breslow thickness) correlates with risk of metastasis. Look for the **ABCDEs**: **A**symmetry, **B**order irregularity, **C**olor variation, **D**iameter > 6 mm, and **E**volution over time. At least 4 different types of melanoma, including superficial spreading **I**, nodular **J**, lentigo maligna **K**, and acral lentiginous **L**. Often driven by activating mutation in BRAF kinase. Primary treatment is excision with appropriately wide margins. Metastatic or unresectable melanoma in patients with BRAF V600E mutation may benefit from vemurafenib, a BRAF kinase inhibitor.



► MUSCULOSKELETAL, SKIN, AND CONNECTIVE TISSUE—PHARMACOLOGY

Arachidonic acid pathway



LTB₄ is a **neutrophil** chemotactic agent.

PGI₂ inhibits platelet aggregation and promotes vasodilation.

Neutrophils arrive “B4” others.

Platelet-Gathering Inhibitor.

Acetaminophen

MECHANISM	Reversibly inhibits cyclooxygenase, mostly in CNS. Inactivated peripherally.
CLINICAL USE	Antipyretic, analgesic, but not anti-inflammatory. Used instead of aspirin to avoid Reye syndrome in children with viral infection.
ADVERSE EFFECTS	Overdose produces hepatic necrosis; acetaminophen metabolite (NAPQI) depletes glutathione and forms toxic tissue byproducts in liver. N-acetylcysteine is antidote—regenerates glutathione.

Aspirin

MECHANISM	NSAID that irreversibly inhibits cyclooxygenase (both COX-1 and COX-2) by covalent acetylation → ↓ synthesis of TXA ₂ and prostaglandins. ↑ bleeding time. No effect on PT, PTT. Effect lasts until new platelets are produced.
CLINICAL USE	Low dose (< 300 mg/day): ↓ platelet aggregation. Intermediate dose (300–2400 mg/day): antipyretic and analgesic. High dose (2400–4000 mg/day): anti-inflammatory.
ADVERSE EFFECTS	Gastric ulceration, tinnitus (CN VIII). Chronic use can lead to acute renal failure, interstitial nephritis, GI bleeding. Risk of Reye syndrome in children treated with aspirin for viral infection. Toxic doses cause respiratory alkalosis early, but transitions to mixed metabolic acidosis-respiratory alkalosis.

Celecoxib

MECHANISM	Reversibly and selectively inhibits the cyclooxygenase (COX) isoform 2 (“ Selecoxib ”), which is found in inflammatory cells and vascular endothelium and mediates inflammation and pain; spares COX-1, which helps maintain gastric mucosa. Thus, does not have the corrosive effects of other NSAIDs on the GI lining. Spares platelet function as TXA ₂ production is dependent on COX-1.
CLINICAL USE	Rheumatoid arthritis, osteoarthritis.
ADVERSE EFFECTS	↑ risk of thrombosis. Sulfa allergy.

Nonsteroidal anti-inflammatory drugs

Ibuprofen, naproxen, indomethacin, ketorolac, diclofenac, meloxicam, piroxicam.

MECHANISM	Reversibly inhibit cyclooxygenase (both COX-1 and COX-2). Block prostaglandin synthesis.
CLINICAL USE	Antipyretic, analgesic, anti-inflammatory. Indomethacin is used to close a PDA.
ADVERSE EFFECTS	Interstitial nephritis, gastric ulcer (prostaglandins protect gastric mucosa), renal ischemia (prostaglandins vasodilate afferent arteriole), aplastic anemia.

Leflunomide

MECHANISM	Reversibly inhibits dihydroorotate dehydrogenase, preventing pyrimidine synthesis. Suppresses T-cell proliferation.
CLINICAL USE	Rheumatoid arthritis, psoriatic arthritis.
ADVERSE EFFECTS	Diarrhea, hypertension, hepatotoxicity, teratogenicity.

Bisphosphonates

Alendronate, ibandronate, risedronate, zoledronate.

MECHANISM	Pyrophosphate analogs; bind hydroxyapatite in bone, inhibiting osteoclast activity.
CLINICAL USE	Osteoporosis, hypercalcemia, Paget disease of bone, metastatic bone disease, osteogenesis imperfecta.
ADVERSE EFFECTS	Esophagitis (if taken orally, patients are advised to take with water and remain upright for 30 minutes), osteonecrosis of jaw, atypical stress fractures.

Teriparatide

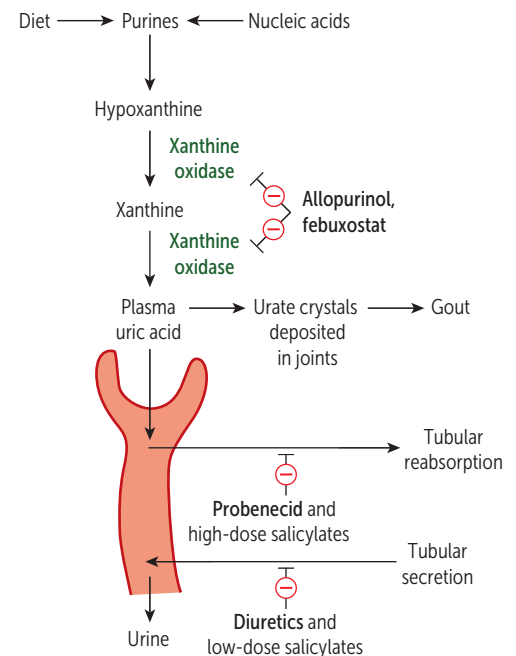
MECHANISM	Recombinant PTH analog given subcutaneously daily. ↑ osteoblastic activity.
CLINICAL USE	Osteoporosis. Causes ↑ bone growth compared to antiresorptive therapies (eg, bisphosphonates).
ADVERSE EFFECTS	↑ risk of osteosarcoma (avoid use in patients with Paget disease of the bone or unexplained elevation of alkaline phosphatase). Avoid in patients who have had prior cancers or radiation therapy. Transient hypercalcemia.

Gout drugs**Chronic gout drugs (preventive)**

Allopurinol	Competitive inhibitor of xanthine oxidase. ↓ conversion of hypoxanthine and xanthine to urate. Also used in lymphoma and leukemia to prevent tumor lysis–associated urate nephropathy. ↑ concentrations of azathioprine and 6-MP (both normally metabolized by xanthine oxidase).
Febuxostat	Inhibits xanthine oxidase.
Pegloticase	Recombinant uricase that catalyzes metabolism of uric acid to allantoin (a more water-soluble product).
Probenecid	Inhibits reabsorption of uric acid in proximal convoluted tubule (also inhibits secretion of penicillin). Can precipitate uric acid calculi.

Acute gout drugs

NSAIDs	Any full-dose NSAID (eg, naproxen, indomethacin). Avoid salicylates (may decrease uric acid excretion, particularly at low doses).
Glucocorticoids	Oral, intra-articular, or parenteral.
Colchicine	Binds and stabilizes tubulin to inhibit microtubule polymerization, impairing neutrophil chemotaxis and degranulation. Acute and prophylactic value. GI side effects.

**TNF- α inhibitors**

All TNF- α inhibitors predispose to infection, including reactivation of latent TB, since TNF is important in granuloma formation and stabilization.

DRUG	MECHANISM	CLINICAL USE
Etanercept	Fusion protein (receptor for TNF- α + IgG ₁ Fc), produced by recombinant DNA. Etanercept intercepts TNF.	Rheumatoid arthritis, psoriasis, ankylosing spondylitis
Infliximab, adalimumab, certolizumab, golimumab	Anti-TNF- α monoclonal antibody.	Inflammatory bowel disease, rheumatoid arthritis, ankylosing spondylitis, psoriasis

▶ NOTES